

Fast Probabilistic Algorithms for Proving Pairings

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Abstract

In this paper, we introduce a public-coin interactive proof system for efficiently proving the correctness of elliptic curve pairing computations. Elliptic curve pairings form the mathematical foundation of many modern cryptographic protocols, yet their verification remains computationally intensive.

In resource-constrained environments—including BitcoinVM, blockchain smart contracts, ZKVMs, and recursive zero-knowledge proof systems—traditional pairing verification remains expensive. This limits the applicability and affordability of privacy-preserving applications, effectively hindering widespread adoption of blockchain technology.

We systematically present a progression of state-of-the-art techniques: El Housni’s foundational R1CS pairing optimizations [Hou22], Feltroidprime’s enhanced extension field arithmetic [Fel23], and Novakovic and Eagen’s final exponentiation elimination [NE24].

Building upon these foundations, our approach consolidates Miller loop iterations and final exponentiation verification into univariate polynomial relations that can be verified through efficient polynomial identity testing. By verifying these polynomial relationships instead of performing the computations we achieve a 50% improvement in Pairing performance. While our method requires additional auxiliary data compared to traditional pairing verification, it achieves a 75% reduction in commitment overhead relative to Feltroidprime’s extension field techniques.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Our Contribution	3
1.3	Paper organization	4
2	Background	4
2.1	Pairing Computation	4
2.2	Schwartz–Zippel lemma	5
2.3	Constraint-Based Computation Models	5

3	Related Work	6
3.1	Pairings in R1CS – Housni 2022	6
3.1.1	Cost of inversions	6
3.1.2	Affine coordinates	6
3.1.3	Avoid doubling	7
3.1.4	Line evaluations	7
3.1.5	Performance	7
3.2	Faster Extension Field Multiplications – Feltroidprime 2023	7
3.2.1	Isomorphisms among towered to direct extensions	7
3.2.2	Multiplication and Schwartz–Zippel lemma	8
3.2.3	Deferred Evaluation and Batching	8
3.2.4	Performance	8
3.3	Eliminating Final Exponentiation – Novakovic & Eagen 2024	9
3.3.1	Equivalence class	9
3.3.2	Equivalence via Residue Check	9
3.3.3	Hasse bound and exponentiation by r	10
3.3.4	Parameters for BN254 curve	10
3.3.5	Performance	10
4	Pairing Operations as Polynomials	11
4.1	Strategic Polynomial Segmentation	11
4.1.1	Squaring operations and polynomial degree	11
4.1.2	Natural segmentation points	11
4.2	Unified Polynomial Framework	13
4.3	Comparative Analysis	13
4.3.1	Previous scheme	13
4.3.2	Our scheme	13
4.3.3	Performance Comparison	14
4.4	Operation Polynomials	14
4.4.1	Miller loop: Zero Bit Equation	14
4.4.2	Miller loop: Non-Zero Bit Equation	14
4.4.3	Miller loop: Frobenius mapping Equation	15
4.4.4	Final Exponentiation Equation	16
4.5	Security	16
4.5.1	Polynomial Identity Testing	16
4.5.2	Euclidean Division	16
4.5.3	Security Analysis	17
5	Interactive Oracle Proof	17
5.1	Batching Polynomial Identity Testing	17
5.2	Prover	17
5.3	Verifier	18
5.4	Performance	18
5.5	[wip] Protocol Overview	18
5.6	[wip] Single-Round Protocol	18
5.6.1	[wip] Protocol description (Algorithm)	18
5.6.2	[wip] Complexity analysis	18
5.7	Two-Round Protocol with RLC Batching	18

5.7.1	[wip] Protocol description (Algorithm)	18
5.7.2	[wip] Complexity analysis	18
5.7.3	[wip] Comparison with single-round	18
5.8	Three-Round Protocol with Native Field Optimization	18
5.8.1	[wip] Protocol description (Algorithm)	18
5.8.2	[wip] Complexity analysis	18
5.8.3	[wip] Architecture-specific benefits	18

List of Algorithms

1	Optimal Ate pairing over Barreto–Naehrig curves [BDM ⁺ 10]	5
2	Optimal Ate pairing with Residue witness check	12

List of Tables

1	Arithmetic in Housni’s paper [Hou22]	7
2	Arithmetic in Feltroidprime’s paper [Fel23]	9
3	Performance impact of residue witness check [NE24]	10
4	Miller loop bit operation comparison (3-pair)	14
5	Zero bit operation polynomial components	15
6	Non-zero bit operation polynomial components	15
7	Correction step polynomial components	15
8	Final exponentiation polynomial components	16

1 Introduction

Blockchain technology is ready for widespread adoption—we have reached a scale where processing Visa-like volumes on-chain is within reach. However, privacy remains a significant barrier to entry for many users and applications. Zero-knowledge proofs (ZKPs) have emerged as a powerful solution to this challenge.

1.1 Motivation

Elliptic curve pairings represent fundamental building blocks for numerous cryptographic systems including Groth16 [Gro16], PLONK [GWC19], BLS signature schemes [BGW⁺22], and KZG polynomial commitment schemes [KZG10].

However, **pairings are computationally intensive**. In execution environments such as blockchain virtual machines, recursive proof systems, and arithmetized circuits, the costs associated with this computation become prohibitive for practical deployment.

1.2 Our Contribution

We present a novel optimization for pairing computation through a public-coin interactive proof system that consolidates the pairing operations into unified polynomial verifications under the Schwartz-Zippel lemma. By treating complete loop iterations as single polynomial relationships rather than sequences of individual field operations [Fel23], our approach achieves:

- **50% reduction** in the pairing computation complexity
- **Over 70% decrease** in auxiliary data overhead
- Practical deployment enabling ZK proof verification within blockchain transaction limits

Our key insight is that instead of optimizing individual field operations, we can treat complete Miller loop iterations as single polynomial relationships, dramatically reducing both computational and communication costs.

Implementation and Impact: Garaga [FE], a state-of-the-art elliptic curve tooling library for Cairo and Starknet, implements our two round interactive proof system with polynomial batching. These techniques allowed Garaga to fit their pairing transaction within Starknet’s 4000 calldata limit while minimising the gas costs.

1.3 Paper organization

The remainder of this paper is organized as follows.

Section 2 provides necessary background on pairings and constraint systems.

Section 3 surveys the evolution of optimization techniques that form our foundation.

Section 4 presents our unified Miller loop optimization framework.

Section 5 outlines the public-coin interactive proof systems with one, two and three rounds for different applications.

Section 6 concludes with implications and future work.

2 Background

2.1 Pairing Computation

An elliptic curve pairing is a non-degenerate bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, where $\mathbb{G}_1$ and $\mathbb{G}_2$ are groups of points on elliptic curves over finite fields, and $\mathbb{G}_T$ is a multiplicative group in an extension field. For practical cryptographic applications, we require that this map be efficiently computable.

The computation of an optimal ate pairing—the focus of our implementation—consists of two primary phases:

1. **Miller loop:** The core iterative computation that evaluates line functions at pairs of points from $\mathbb{G}_1$ and $\mathbb{G}_2$, accumulating intermediate results to produce a value in $\mathbb{F}_{q^{12}}$.
2. **Final Exponentiation:** This step raises the Miller loop output to the power $(q^{12} - 1)/r$, where r is the order of the groups, ensuring the result lies in the correct subgroup $\mathbb{G}_T$.

While both phases present optimization opportunities, the constraint-based verification of these computations introduces unique challenges and possibilities that differ significantly from native implementations.

Algorithm 1 Optimal Ate pairing over Barreto–Naehrig curves [BDM⁺10]

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$ **Output:** $a_{\text{opt}}(Q, P)$

```
1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $T \leftarrow Q, f \leftarrow 1$ 
3: for  $i = L - 2$  to  $0$  do
4:    $f \leftarrow f^2 \cdot l_{T,T}(P)$ 
5:    $T \leftarrow [2]T$ 
6:   if  $s_i = 1$  then
7:      $f \leftarrow f \cdot l_{T,Q}(P)$ 
8:      $T \leftarrow T + Q$ 
9:   end if
10:  if  $s_i = -1$  then
11:     $f \leftarrow f \cdot l_{T,-Q}(P)$ 
12:     $T \leftarrow T - Q$ 
13:  end if
14: end for
15:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 
16:  $f \leftarrow f \cdot l_{T,Q_1}(P), T \leftarrow T + Q_1$ 
17:  $f \leftarrow f \cdot l_{T,-Q_2}(P), T \leftarrow T - Q_2$ 
18:  $f \leftarrow f^{p^{12}-1/r}$ 
19: return  $f$ 
```

2.2 Schwartz–Zippel lemma

The Schwartz–Zippel lemma (or more precisely, the Demillo–Lipton–Schwartz–Zippel lemma [DL78, Sch79, Zip79, Sch80]) provides a probabilistic method for testing polynomial identities. The lemma states that for a non-zero polynomial $P(x_1, x_2, \dots, x_n)$ of total degree d over a finite field $\mathbb{F}$, when evaluated at uniformly random field elements, the probability that P evaluates to zero is at most $d/|\mathbb{F}|$.

This probabilistic bound becomes negligible for cryptographically sized fields where $d \ll |\mathbb{F}|$, making it practical to verify polynomial identities by evaluation instead of coefficient-wise operations.

2.3 Constraint-Based Computation Models

In the context of arithmetization protocols and virtual machines (such as Starkware’s Cairo VM [Sta22], R1CS [GGPR13], Plonkish [GWC19], etc.), where field operations are expressed as algebraic constraints, the computational landscape differs dramatically from native implementations. Operations that are traditionally expensive can become remarkably efficient. Field inversions, for example, can be computed at a fraction of their usual cost, often reducing their relative costs from over 100× to about the same cost as multiplication. Conversely, seemingly straightforward operations may require complex constraint representations.

Recognizing and exploiting these unique computational properties is essential for unlocking hidden performance potential. Youssef El Housni’s pioneering work **Pairings in Rank-1 Constraint Systems** [Hou22] represents, to our knowledge, the first systematic exploration of these optimization opportunities, providing a breakthrough in the practi-

quality of recursive proving by thoughtfully leveraging the distinctive cost model inherent to constraint-based systems.

3 Related Work

The landscape of efficient pairing verification in constraint systems has evolved through several key contributions. El Housni’s **Pairings in Rank-1 Constraint Systems** [Hou22] established the theoretical framework for optimizing pairings in Rank-1 Constraint Systems, introducing techniques for efficient Miller loop implementation and final exponentiation that exploit the unique properties of constraint-based computation models. This work demonstrated that operations traditionally considered expensive in native implementations could be restructured to achieve remarkable efficiency gains in arithmetized environments.

Feltroidprime’s subsequent research on **enhanced extension field multiplication techniques** [Fel23] introduced polynomial verification methods using the Schwartz-Zippel lemma for $\mathbb{F}_{q^{12}}$ arithmetic operations. By representing extension field elements as polynomials and verifying their operations through polynomial arithmetic, this approach achieved significant performance improvements in the pairing computations, particularly for emulated pairing circuits where traditional field operations are costly.

Novakovic and Eagen’s work **On Proving Pairings** [NE24] presented breakthrough optimization strategies that enable the elimination of final exponentiation overhead through residue witness techniques. Their approach demonstrates that two elements in $\mathbb{F}_{q^{12}}$ can be proven equivalent without expensive exponentiation by introducing witness elements that satisfy specific algebraic relationships, effectively embedding the final exponentiation verification into the Miller loop computation itself.

3.1 Pairings in R1CS – Housni 2022

El Housni’s seminal work [Hou22] represents the first systematic exploration of pairing optimization within constraint-based computation models, providing techniques that extend beyond R1CS to benefit any arithmetized environment, including Starkware’s Cairo VM [Sta22].

3.1.1 Cost of inversions

Housni notes that the cost of inversions is much lower in constraint systems, almost the same as multiplication. Over $\mathbb{F}_p$, inversions cost 2 constraints: one for computing $x \cdot y$ where y is provided from the hint, and one equality check $x \cdot y \stackrel{?}{=} 1$. Division can be similarly reduced to just two constraints. For $c = a/b$ where c is the input from hint, we compute $b \cdot c$ and equality check $b \cdot c \stackrel{?}{=} a$. This is further generalised to field extensions.

3.1.2 Affine coordinates

Since division cost about as much as multiplications, we can represent points in affine coordinates for lower operation counts. Point doubling and addition formulas are provided in section 6.1 of [Hou22]. Doubling costs 12C and addition costs 10C.

3.1.3 Avoid doubling

For double and add operations on non zero bits of the Miller loop, instead of computing $[2]S + Q$, computing $S + Q + S$ saves 2C per non zero bit. Taking this a step further, computation of y coordinate for the intermediate $S + Q$ point can be omitted and λ_1 can be directly used in computation of λ_2 . This **costs 17C**, that’s a **23% savings** over 22C for $[2]S + Q$.

3.1.4 Line evaluations

Following [ABLR14], the lines are sparse elements in $\mathbb{F}_{q^{12}}$ of the form $ay_P + bx_P \cdot w + c \cdot w^3$ with $a, b, c \in \mathbb{F}_{p^2}$. The multiplication of these sparse elements costs 39C. Housni proposes representing the lines in the form $1 + \frac{b'x_P}{y_P} \cdot w + \frac{c'}{y_P} \cdot w^3$ which makes it sparser (notice unit first coefficient resulting in free multiplication).

Terms $\frac{x_P}{y_P}$ and $\frac{1}{y_P}$ can be precomputed at the cost of 2C, and reused throughout the miller loop. Untwisting isomorphisms for this form are given on pages 15–16 in section 6.1 of [Hou22]. This representation only **costs 30C**, that’s a **30% savings** over 39C for the original representation.

3.1.5 Performance

Table 1 summarizes the arithmetic costs for $\mathbb{F}_{q^{12}}$ as presented in Housni’s paper.

Table 1: Arithmetic in Housni’s paper [Hou22]

	Mul	Square	Div	Sparse Mul
$\mathbb{F}_{q^{12}}$	54C	36C	66C	30C

3.2 Faster Extension Field Multiplications – Feltroidprime 2023

Feltroidprime [Fel23] showcases a novel approach to extension field multiplications that exploits a counterintuitive property of circuit-based computation: operations traditionally viewed as expensive (such as polynomial divisions) can become highly efficient in constraint systems. Moving away from the conventional towered extension approach, he proposes returning to direct extensions of $\mathbb{F}_p$ to achieve significant performance improvements.

3.2.1 Isomorphisms among towered to direct extensions

Feltroidprime walks through the construction of mapping functions to go back and forth between direct and towered extensions.

He starts with the tower for BN254 (as used in gnark [Con]) and obtains irreducible polynomial $-P_{12}(x) = x^{12} - 18x^6 + 82$

And prepares isomorphisms between tower and direct representations [Fel] which we will not include in this paper for brevity.

3.2.2 Multiplication and Schwartz–Zippel lemma

Feltroidprime proposes using Euclidean division of polynomials to multiply $A, B \in \mathbb{F}_{q^{12}}$. The computation happens as follows:

1. Treat A, B as polynomials of degree less than 11, with coefficients in $\mathbb{F}_p$.
2. Compute the product $C = A \cdot B$ as a polynomial of degree less than 22 for $\mathbb{F}_{q^{12}}$.
3. Perform Euclidean division of C by the irreducible polynomial $P_{12}(x)$ to obtain the quotient Q and remainder R : $\boxed{A(x) \cdot B(x) = Q(x) \cdot P_{12}(x) + R(x)}$
4. $R \in \mathbb{F}_{q^{12}}$ is used as the result of the multiplication between A and B .

Inside the circuit, this computation can be provided as a hint and verified with the Schwartz–Zippel lemma. The Fiat–Shamir heuristic can be used to generate the challenge with the transcript.

3.2.3 Deferred Evaluation and Batching

In Section 4, Feltroidprime discusses deferring the evaluation to the end of the algorithm and combining it with a polynomial commitment scheme to batch the verifications. This batching transforms hundreds of individual verifications into a single equation. So they can be checked at a single random point, amortizing the costs for Fiat-Shamir hashing.

The batching process is a random linear combination of all the equations to be verified. For n multiplications, we have:

$$\sum_{i=0}^{n-1} c_i * A_i(z) * B_i(z) = \sum_{i=0}^{n-1} c_i * R_i(x) + \sum_{i=0}^{n-1} c_i * Q_i(z) * P_{12}(z) \quad (1)$$

3.2.4 Performance

For performance analysis, we will use the batched verification version.

The implementation receives the computed RHS polynomial aggregation $Q_{Agg} = \sum_{i=0}^{n-1} c_i * Q_i$ as hint. The LHS aggregation, $\sum_{i=0}^{n-1} c_i * A_i(z) * B_i(z) - R_i(x)$ is computed within the verifier circuit. While the amortized cost computing the $Q_{Agg}(z)$ and $P_{12}(z)$ once in the circuit approaches zero as the number of batched operations increases, we conservatively account for 1C per multiplication to provide fair cost comparison.

Multiplication costs 12C each for A, B and R evaluation, 1C for multiplying $A(z)$ and $B(z)$ evaluations, plus 1C for preparing the RLC c_i multiplication and conservative 1C for RHS aggregation.

For **squarings**, the evaluation of $A(z)$ is just used again, costing 12C less than multiplication.

Division costs multiplication plus additional 12C for assertion.

Sparse element multiplication costs Evaluations for A, B and R where B contains 5 non-zero coefficients, one out of which is unitary free multiplication. The costs comes at 31C which is 4C more than squaring which has free second polynomial because $A = B$ for squarings.

Table 2: Arithmetic in Feltroidprime’s paper [Fel23]

	Mul	Square	Div	Sparse Mul
$\mathbb{F}_{q^{12}}$	39C*	27C*	51C*	31C*
*Costs exclude Fiat-Shamir commitment overhead				

3.3 Eliminating Final Exponentiation – Novakovic & Eagen 2024

The mathematical structure underlying pairing based cryptography involves equivalence classes. Tate pairings (as described in [NE24] and [Cos19]) result in the output in the quotient group $F_{q^k}/(F_{q^k})^r$. This requires the output of the Miller loop to be raised to a power of $(q^k - 1)/r$. This step, called **the final exponentiation**, normalizes the Miller loop output by clearing the r -th residue part. This normalization ensures equivalent pairing outputs produce identical results.

This operation is particularly expensive in extension fields like $\mathbb{F}_{q^{12}}$, because $(q^{12} - 1)/r$ becomes very large.

3.3.1 Equivalence class

Novakovic and Eagen [NE24] highlight that the final exponentiation can be eliminated when proving pairings. The expensive exponentiation can be replaced with an efficient residue check. The key insight is that instead of normalizing pairing outputs through final exponentiation, we can directly verify equivalence using a residue-witness based approach.

Equivalence class representation: Two elements $A, B \in \mathbb{F}_{q^{12}}$ are equivalent in the pairing context if there exists some witness c such that $A \cdot c^r = B$.

$$A \equiv B \iff \exists c \in \mathbb{F}_{q^{12}} : A \cdot c^r = B$$

3.3.2 Equivalence via Residue Check

So we have,

$$A \cdot c^r \equiv A$$

After final exponentiation,

$$\begin{aligned} (A \cdot c^r)^{(q^{12}-1)/r} &\equiv A^{(q^k-1)/r} \\ A^{(q^k-1)/r} \cdot c^{r \cdot (q^k-1)/r} &\equiv A^{(q^k-1)/r} \\ A^{(q^k-1)/r} \cdot c^{q^k-1} &\equiv A^{(q^k-1)/r} \end{aligned}$$

With Fermat’s Little Theorem,

$$A^{(q^k-1)/r} \cdot 1 \equiv A^{(q^k-1)/r}$$

If we expect the result of the pairing, $A \in \mathbb{F}_{q^k}$, to be equal to some desired output $B \in \mathbb{F}_{q^k}$, we can use the equivalence relation to establish there exists some $c \in \mathbb{F}_{q^k}$ such that $A \cdot c^r = B$.

This can be generalized to $A \cdot w^{rt} = B$ for $c = w^t$ in $A \cdot c^r = B$.

3.3.3 Hasse bound and exponentiation by r

According to Hasse theorem [Sil09], $q + 1 - r \leq 2\sqrt{q}$ for an elliptic curve over finite field $\mathbb{F}_q$ and r the order of the curve.

The exponentiation by r can be written as exponentiation by $q + 1 - t$. Where $t \leq 2\sqrt{q}$ is much smaller exponentiation in worst case. And the computation of c^q is efficiently computed by exploiting the Frobenius endomorphisms.

3.3.4 Parameters for BN254 curve

Section 4.2 and 4.3 of [NE24] dive into finding optimal parameters for the curve to establish $A \cdot c^\lambda = B$

For Ethereum’s BN254 curve, the optimal parameter is $\lambda = 6x + 2 + q - q^2 + q^3$, which decomposes into two efficiently computable components:

1. **Miller loop component** $(6x + 2)$ — Integrated into existing Miller loop operations
2. **Frobenius component** $(q - q^2 + q^3)$ — Computed using efficient Frobenius mappings

This eliminates the expensive final exponentiation, replacing it with a few additional Miller loop operations and three Frobenius mappings.

3.3.5 Performance

Replacing the final exponentiation by residue witness check saves over 85% of final exponentiation costs for BN254 curve, reducing the constraints required for whole pairing by 32%.

Table 3 summarizes the performance impact of eliminating the final exponentiation step, comparing the original pairing costs with those after applying the residue witness technique.

Table 3: Performance impact of residue witness check [NE24]

Computation	Full Pairing	Witness Check [NE24]	Change
Emulated BN254 pairing			
Final Exponentiation	295,823C	38,276C	-87.1%
Total cost	784,050C	526,503C	-32.8%
Native BLS12377 [EHG22] pairing			
Final Exponentiation	6,130C	396C	-93.5%
Total cost	14,536C	8,802C	-39.4%

4 Pairing Operations as Polynomials

The techniques discussed in the previous section are outlined in Algorithm 2. This algorithm combines El Housni’s R1CS optimizations [Hou22] with Novakovic and Eagen’s residue witness technique for final exponentiation elimination [NE24].

We now turn our attention to our primary contribution. Building upon Feltroidprime’s direct extension multiplication framework, we present a novel optimization that significantly improves efficiency across all phases of the pairing computation. Our approach consolidates operations across the entire process into unified polynomial verifications. This includes the Miller loop step operations as well as the final exponentiation (via residue witness) verification. Each phase is expressed as a polynomial relationship and verified using the Schwartz-Zippel lemma, substantially reducing constraint count while maintaining cryptographic security.

4.1 Strategic Polynomial Segmentation

A critical consideration in our design is managing polynomial degree growth. While consolidating operations into fewer polynomials reduces overhead, we must balance this against polynomial degree constraints to maintain verification efficiency.

4.1.1 Squaring operations and polynomial degree

Squaring operations **double the polynomial degree**. In traditional $\mathbb{F}_{q^{12}}$ arithmetic, squaring operations are typically cheaper than general multiplications. However, in our polynomial verification framework, squaring the accumulator F doubles the polynomial degree, making placement critical. When positioned after other operations, squaring becomes the most degree-expensive operation in the sequence.

4.1.2 Natural segmentation points

The optimal points to segment our unified polynomial are immediately *before* squaring the accumulator. By making squaring the first operation in each polynomial segment, we:

1. Minimize the degree growth impact—the squaring happens on a fresh polynomial of degree 11
2. Accumulate all subsequent multiplications at manageable degree increases
3. Maintain relatively uniform polynomial degrees across segments

This observation naturally leads to segmenting operations at each Miller loop step boundary, where the accumulator F is squared. The remaining operations are then segmented to match the degree profile of Miller loop step polynomials, ensuring balanced verification costs across all phases.

Algorithm 2 Optimal Ate pairing with Residue witness check

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$ Internal $W, W^2 \in \mathbb{F}_{q^{12}}$ $\triangleright W$ is 27th root of unity**Output:** $a_{opt}(Q, P)$ 1: Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$ 2: $c^{-1} \leftarrow 1/c$ $\triangleright$ Inverse Residue witness3: $f \leftarrow c^{-1}$ $\triangleright$ Initialize with inverse residue witness4: $T \leftarrow Q$ $\triangleright T_i$ for each pair for multi-pairingEvery operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.5: **for** $i = L - 2$ to 0 **do**6: $f \leftarrow f^2$ 7: $f \leftarrow f \cdot l_{T,T}(P)$ $\triangleright$ Repeat for T_i, P_i 8: $T \leftarrow [2]T$ $\triangleright$ Repeat for T_i 9: **if** $s_i = 1$ **then**10: $f \leftarrow f \cdot c^{-1}$ $\triangleright$ Residue witness11: $f \leftarrow f \cdot l_{T,Q}(P)$ $\triangleright$ Repeat for T_i, Q_i, P_i 12: $T \leftarrow T + Q$ $\triangleright$ Repeat for T_i, Q_i 13: **else if** $s_i = -1$ **then**14: $f \leftarrow f \cdot c$ $\triangleright$ Residue witness15: $f \leftarrow f \cdot l_{T,-Q}(P)$ $\triangleright$ Repeat for $T_i, -Q_i, P_i$ 16: $T \leftarrow T - Q$ $\triangleright$ Repeat for $T_i, -Q_i$ 17: **end if**18: **end for**19: $Q' \leftarrow \pi_p(Q)$ $\triangleright$ Repeat for Q'_i, Q_i 20: $f \leftarrow f \cdot l_{T,Q'}(P)$ $\triangleright$ Repeat for Q'_i, T_i, P_i 21: $T \leftarrow T + Q'$ $\triangleright$ Repeat for Q'_i, T_i 22: $f \leftarrow f \cdot l_{T,-\pi_p^2(Q)}(P)$ $\triangleright$ Repeat for Q_i, T_i, P_i $\triangleright$ Skip $T \leftarrow T + -\pi_p^2(Q)$ 23: **if** $w = 1$ **then**24: $f \leftarrow f \cdot W$ 25: **else if** $w = 2$ **then**26: $f \leftarrow f \cdot W^2$ 27: **end if**28: $f \leftarrow f \cdot c^{-q} \cdot c^{q^2} \cdot c^{-q^3}$ $\triangleright$ Uses Frobenius maps, Negative exponents use c^{-1} 29: **return** f

4.2 Unified Polynomial Framework

Any sequence of field operations in $\mathbb{F}_{q^{12}}$ ¹ can be expressed as a single polynomial equation of the form:

$$A_1(x) \cdot \dots \cdot A_n(x) = Q(x) \cdot P_{12}(x) + R(x) \text{ where } A_1, \dots, A_n, R \in \mathbb{F}_{q^{12}}$$

This unified framework allows us to batch verify entire computational phases rather than individual operations, achieving dramatic improvements in both constraint counts and commitment overhead across the complete pairing computation.

4.3 Comparative Analysis

The Miller loop is comprised of steps for each individual bit, which in NAF (non-adjacent form) [HMOV04], may be +1, -1 or 0. Instead of performing a polynomial commitment for each individual extension field arithmetic operation separately, we propose the whole bit operation be viewed as one polynomial. And commitment be made to that polynomial.

To get a closer look into effectiveness of our scheme, we will make a real world application comparison for verification of three-pairing Miller loop which is common for verification of Groth16 [Gro16]. We benchmark our approach against the fastest known polynomial commitment adaptation [Fel23].

4.3.1 Previous scheme

Adapting faster extension field multiplication [Fel23] idea to line 6 and 7 of Miller loop algorithm 2 for a 3-pair Miller loop, we have following polynomial relationships to assert.

$$\begin{aligned} F(x) \cdot F(x) &\stackrel{?}{=} Q_1(x) \cdot P_{12}(x) + T_1(x) \\ T_1(x) \cdot L_1(x) &\stackrel{?}{=} Q_2(x) \cdot P_{12}(x) + T_2(x) \\ T_2(x) \cdot L_2(x) &\stackrel{?}{=} Q_3(x) \cdot P_{12}(x) + T_3(x) \\ T_3(x) \cdot L_3(x) &\stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x) \end{aligned}$$

F, T_1, T_2, T_3, R cost 12C each for evaluation, 4C for $\mathbb{F}_q$ multiplication, and L_1, L_2, L_3 require 4C each for evaluation. All the remainder hints need to be provided for commitment, so this is 12 calldata commitment for R and intermediate remainder polynomials T_1, T_2, T_3 . **Costs sum up to 76C ($5 * 12C + 3 * 4C + 4C$) and 48 commitments** in $\mathbb{F}_q$ ($4 * 12$). Resulting remainder polynomial $R \in \mathbb{F}_{q^{12}}$ is the result of the Miller loop step.

4.3.2 Our scheme

In our approach, instead of working with individual arithmetic operations, we treat the entire bit operation as a single polynomial. In our scheme line 6 and 7 of 3-pair Miller loop (Algorithm 2) looks like this.

¹Elements need not have all 12 non-zero coefficients, but should align with the $\mathbb{F}_{q^{12}}$ representation (e.g., sparse line functions).

$$F(x) \cdot F(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)$$

Similar to the previous approach, F and R cost 12C each for evaluation, 4C for $\mathbb{F}_q$ multiplication, and L_1, L_2, L_3 require 4C each for evaluation, but we eliminate intermediate (T_* polynomials above) commitments and evaluations. Only the remainder hint needs commitment, so this is 12 calldata commitment for R . **Costs sum up to 40C** ($2 * 12C + 3 * 4C + 4C$) **and 12 $\mathbb{F}_q$ commitments.**

This results in **47% fewer constraints** and a whopping **75% savings in the commitments**. When implemented on a blockchains for verification of ZK proofs, commitment savings become far more critical because of limitations and costs associated with calldata needed for hints.

4.3.3 Performance Comparison

Table 4 compares the computational costs and commitment overhead between the schemes for a single 0-bit Miller loop operation with 3 pairs.

Table 4: Miller loop bit operation comparison (3-pair)

Scheme	Constraints	$\mathbb{F}_q$ Commitments	Improvement
[Hou22]	132C	—	Baseline
[Fel23]	76C*	48	42%
Our scheme	40C*	12	70%
[Hou22] (Table 1): 1 square (36C) + 3 sparse mul ($3 \times 30C$) = 132C [Fel23]: 5 evals ($5 \times 12C$) + 3 line evals ($3 \times 4C$) + 4 muls ($4 \times 1C$) = 76C Our scheme: 2 evals ($2 \times 12C$) + 3 line evals ($3 \times 4C$) + 4 muls ($4 \times 1C$) = 40C *Costs exclude Fiat-Shamir commitment overhead			

4.4 Operation Polynomials

Our implementation uses different polynomials for different bits in the Miller loop. Let's walk through a 3-pairing setup which Inversions used widely for Groth16 verification. We will describe the polynomials for zero bits, non-zero bits and correction step.

4.4.1 Miller loop: Zero Bit Equation

As seen in 4.3.2 for zero bit operations (line 6 and 7 with three $l_{T,T}(P)$ lines in Algorithm 2), the verification equation is:

$$F^2(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (2)$$

4.4.2 Miller loop: Non-Zero Bit Equation

For non-zero bit operations (lines 10, 11, 14, 15 with three pairs of lines in Algorithm 2), the Miller loop performs both doubling and addition steps. So for each pair we have two Sparse₀₁₃₇₉ lines. The verification equation becomes:

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_1, L_2, L_3 \in \text{Sparse}_{01379}$	The line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total multiplications			36
Q Amortized	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	38	39

Table 5: Zero bit operation polynomial components

$$F^2(x) \cdot W(x) \cdot L_{d1}(x) \cdot L_{d2}(x) \cdot L_{d3}(x) \cdot L_{a1}(x) \cdot L_{a2}(x) \cdot L_{a3}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (3)$$

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_{d1}, L_{d2}, L_{d3} \in \text{Sparse}_{01379}$	Doubling line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{a1}, L_{a2}, L_{a3} \in \text{Sparse}_{01379}$	Addition line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
W or $W^{-1} \in \mathbb{F}_{q^{12}}$	Residue witness	11	12
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total multiplications			60
Q Amortized	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	76	77

Table 6: Non-zero bit operation polynomial components

The residue witness W encapsulates the equivalence class relationship established by Novakovic and Eagen’s final exponentiation elimination. The inverse of W is used for negative bits, this just changes the coefficients the equation remains the same.

4.4.3 Miller loop: Frobenius mapping Equation

The correction step finalizes the Miller loop by incorporating the Frobenius endomorphism corrections (lines 20 and 22 in Algorithm 2). Again, we have two Sparse_{01379} lines for each pair, for π_p and $-\pi_p^2$ mappings, without any squaring of the accumulator. The verification equation becomes:

$$F(x) \cdot L_{1\pi}(x) \cdot L_{2\pi}(x) \cdot L_{3\pi}(x) \cdot L_{1\pi^2}(x) \cdot L_{2\pi^2}(x) \cdot L_{3\pi^2}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (4)$$

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	11	12
$L_{1\pi}, L_{2\pi}, L_{3\pi} \in \text{Sparse}_{01379}$	π_p correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{1\pi^2}, L_{2\pi^2}, L_{3\pi^2} \in \text{Sparse}_{01379}$	$-\pi_p^2$ correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total coefficients			48
Q Amortized	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	54	55

Table 7: Correction step polynomial components

4.4.4 Final Exponentiation Equation

The final exponentiation check is handled as described in [NE24]. The equation handles the cubic scale factor multiplication, and **Frobenius component** ($q - q^2 + q^3$) of the final exponentiation (lines 24, 26 and 28 in Algorithm 2). The verification equation looks like:

$$F(x) \cdot c^{-q}(x) \cdot c^{q^2}(x) \cdot c^{-q^3}(x) \cdot W^w(x) = 1 + Q(x) \cdot P_{12}(x) \quad (5)$$

where $w \in \{0, 1, 2\}$ determines the cubic root of unity scaling factor, and the Frobenius mappings are computed using the residue witness c and its inverse and provided as input. Cubic root is sparse element with just 2 coefficients the degree ranging from 8 to 10 because of positioning of the coefficients. Our R is now 1 for a successful proof verification.

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Final Miller loop result	11	12
$c^{-q} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q$	11	12
$c^{q^2} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius q^2	11	12
$c^{-q^3} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q^3$	11	12
$W^w \in \mathbb{F}_{q^{12}}$	Cubic scale factor	10	2
$R = 1$	Unity (constant)	0	1
Total coefficients			51
Q Amortized	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	43	44

Table 8: Final exponentiation polynomial components

With this we have all the polynomials we need for verifying entire pairing operation. Next we will look into high level overview of pairing verification via an interactive oracle proof.

4.5 Security

We analyze the security of our polynomial verification scheme.

4.5.1 Polynomial Identity Testing

A polynomial P of degree d over a field $\mathbb{F}_q$ has at most d roots in $\mathbb{F}_q$. Thus for a non-zero univariate polynomial evaluated at a random point $r \xleftarrow{\$} \mathbb{F}_q$, the probability of zero output is,

$$\Pr[P(r) = 0] \leq \frac{d}{|\mathbb{F}_q|}$$

4.5.2 Euclidean Division

Given polynomials A and B in $\mathbb{F}_q[x]$ with $B \neq 0$, there exist unique polynomials Q and R where $\deg(R) < \deg(B)$ such that:

$$A(x) = B(x) \cdot Q(x) + R(x)$$

4.5.3 Security Analysis

Our polynomials are of the form $P = A_1(x) \cdot A_2(x) \cdots A_n(x) - Q(x) \cdot P_{12}(x) - R(x)$ where $\deg(R) \leq 11$ and $\deg(P) < 2^8$. For verifying the computation of R with respect to P_{12} :

Soundness: The soundness error is $\delta_s = \Pr[P(r) = 0] \leq \frac{d}{|\mathbb{F}_q|}$ where $d = \deg(P) < 2^8$. For the 254-bit BN254 field, $|\mathbb{F}_q| \approx 2^{254}$, so $\delta_s \leq 2^{-246}$.

Completeness: If the prover is honest and provides correct Q and R , then $P(x) = 0$ for all $x \in \mathbb{F}_q$. Thus, the verifier always accepts.

5 Interactive Oracle Proof

Now that we have covered the core construction behind our contribution, let's take a holistic view of verification of a pairing computation with an interactive proof system. While the construction above works beautifully on it's own, we can further optimize the protocol by batching all polynomial verifications into a two round interactive proof.

5.1 Batching Polynomial Identity Testing

The whole process of polynomial verification can be batched into a single polynomial verification. Here's how a two round interactive proof system would look like,

The prover can commit all R s for all n equations involved in pairing computation in the first interaction and receive a random challenge z . The challenge z can then be used to batch all the Q s into a single polynomial as a consecutive powers random linear combination like this,

$$Q_{acc} = \sum_{i=0}^{n-1} z^i \cdot Q_i(z)$$

The verifier then prepares a new challenge c and checks,

$$\sum_{i=0}^{n-1} z^i \cdot A_i(c) \cdot B_i(c) \cdots X_i(c) = \sum_{i=0}^{n-1} z_i \cdot R_i(c) + Q_{acc}(c) \cdot P_{12}(c)$$

or,

$$\sum_{i=0}^{n-1} z^i \cdot [A_i(c) \cdot B_i(c) \cdots X_i(c) - R_i(c)] = Q_{acc}(c) \cdot P_{12}(c) \quad (6)$$

This reduces n polynomial evaluations of degree ranging from 38 to 76 (as described in Tables 5, 6, 7, 8) to a single polynomial of degree 76.

5.2 Prover

Prover performs the pairing verification herself, and computes and stores all R_i and Q_i . After the pairing computation, she uses same hashing method as the verifier to obtain all c_i . She then computes the RLC of all Q_i and sends it along with all R_i commitments.

5.3 Verifier

Verifier receives all the remainders, R_i , and RLC of quotients - $\sum_{i=1}^n c_i \cdot Q_i$. The verifier then prepares all c_i solely from the remainder coefficients. For the evaluation point x , she includes all coefficients from the quotient RLC in the Fiat-Shamir hash as well. This guarantees soundness in Fiat Shamir. Then she uses generated c_i for aggregating LHS_i evaluated at x . In the end, she subtracts the evaluation of quotients RLC at x times $P_{12}(x)$ from the aggregated LHS evaluation. If the result is zero, according to Schwartz–Zippel lemma, all the provided R_i are correct with a very high² probability.

5.4 Performance

Batching operations in Miller loop steps yielded substantial performance improvements. Processing all operations within a Miller loop step as a single polynomial and performing Schwartz-Zippel verification of polynomial multiplications significantly reduced the overall computational requirements of the pairing verification process. Combined with residue witness technique for final exponentiation elimination [NE24], these optimizations enabled processing of ZK proofs on Starknet even before the modulo builtin [Sta] became available, which later greatly improved finite field arithmetic operations. These techniques were subsequently integrated into Garaga, which remains the most efficient pairing implementation by operation count on Starknet to date.

5.5 [wip] Protocol Overview

5.6 [wip] Single-Round Protocol

5.6.1 [wip] Protocol description (Algorithm)

5.6.2 [wip] Complexity analysis

5.7 Two-Round Protocol with RLC Batching

5.7.1 [wip] Protocol description (Algorithm)

5.7.2 [wip] Complexity analysis

5.7.3 [wip] Comparison with single-round

5.8 Three-Round Protocol with Native Field Optimization

5.8.1 [wip] Protocol description (Algorithm)

5.8.2 [wip] Complexity analysis

5.8.3 [wip] Architecture-specific benefits

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